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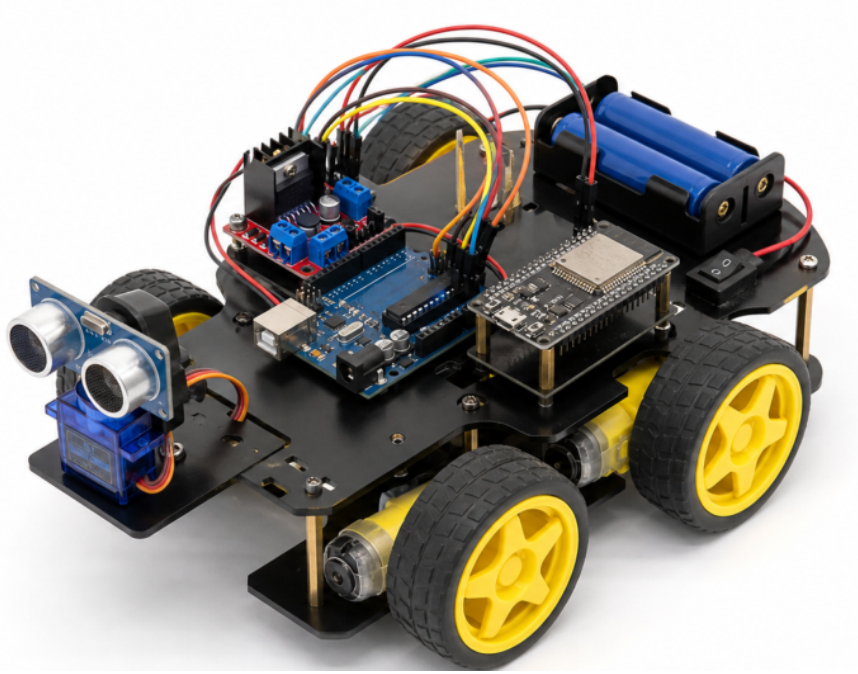


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MINI PROJECT · FINAL REVIEW

IoT-BASED SMART OBSTACLE-AVOIDING ROBOT

Using Arduino UNO and ESP32 for Real-Time Remote Monitoring



PROJECT MEMBERS

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PROJECT GUIDE

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Project Overview

- Low-cost indoor mobile robot combining local obstacle avoidance with IoT telemetry.
- Arduino UNO R3 runs the time-critical sensing, decision and motor-control loop.
- ESP32 provides Wi-Fi connectivity and forwards status data to a ThingSpeak dashboard.
- Remote visibility helps observe obstacle distance, robot state and operational mode.

PROBLEM

Offline robots provide little remote visibility once moving.

SOLUTION

Split local control and cloud telemetry across two controllers.

WHY IT MATTERS

Supports affordable robotics education, indoor inspection prototypes and basic logistics demonstrations.

Scope: indoor reactive navigation; no SLAM or global route planning.

Abstract

Small-scale autonomous robots often lack remote observability or require more expensive computing platforms. This project combines an Arduino UNO R3 for local navigation with an ESP32 telemetry bridge for Wi-Fi monitoring.

An HC-SR04 sensor mounted on an SG90 servo measures front distance and scans left/right when clearance falls below 30 cm. The UNO stops, reverses briefly, compares side clearance and turns toward the clearer path. Telemetry is sent to ThingSpeak at approximately 20-second intervals.

CONTROLLERS

Arduino UNO R3 + ESP32

SENSING & DRIVE

HC-SR04 · SG90 · L293D · 4 TT motors

CLOUD

Wi-Fi telemetry · ThingSpeak dashboard

Low-cost, modular prototype

Literature Review

YEAR	AUTHORS	PAPER / METHOD	RELEVANCE
1991	Borenstein & Koren	Vector Field Histogram (VFH)	Foundation for local reactive steering.
2020	Mothe et al.	IoT-based obstacle avoidance robot	Cloud telemetry with basic avoidance.
2021	Ekta et al.	Ultrasonic obstacle-avoiding vehicle	Indoor ultrasonic sensing.
2024	Katona et al.	Path planning methods review	Context for reactive vs. complex planning.
2024	Aishwarya et al.	Arduino robot using L293D	UNO + HC-SR04 baseline architecture.

Research takeaway: combine a proven reactive controller with an ESP32-based cloud telemetry bridge.

Base Paper & Project Extension

BASE PAPER

"Obstacle avoidance and path planning methods for autonomous navigation of mobile robot"

K. Katona, H. A. Neamah and P. Korondi, Sensors, 2024.

RELATED IMPLEMENTATION BASIS

Project report references reactive navigation, ultrasonic ranging, motor control and IoT telemetry studies.

LIMITATIONS IDENTIFIED IN EXISTING APPROACHES

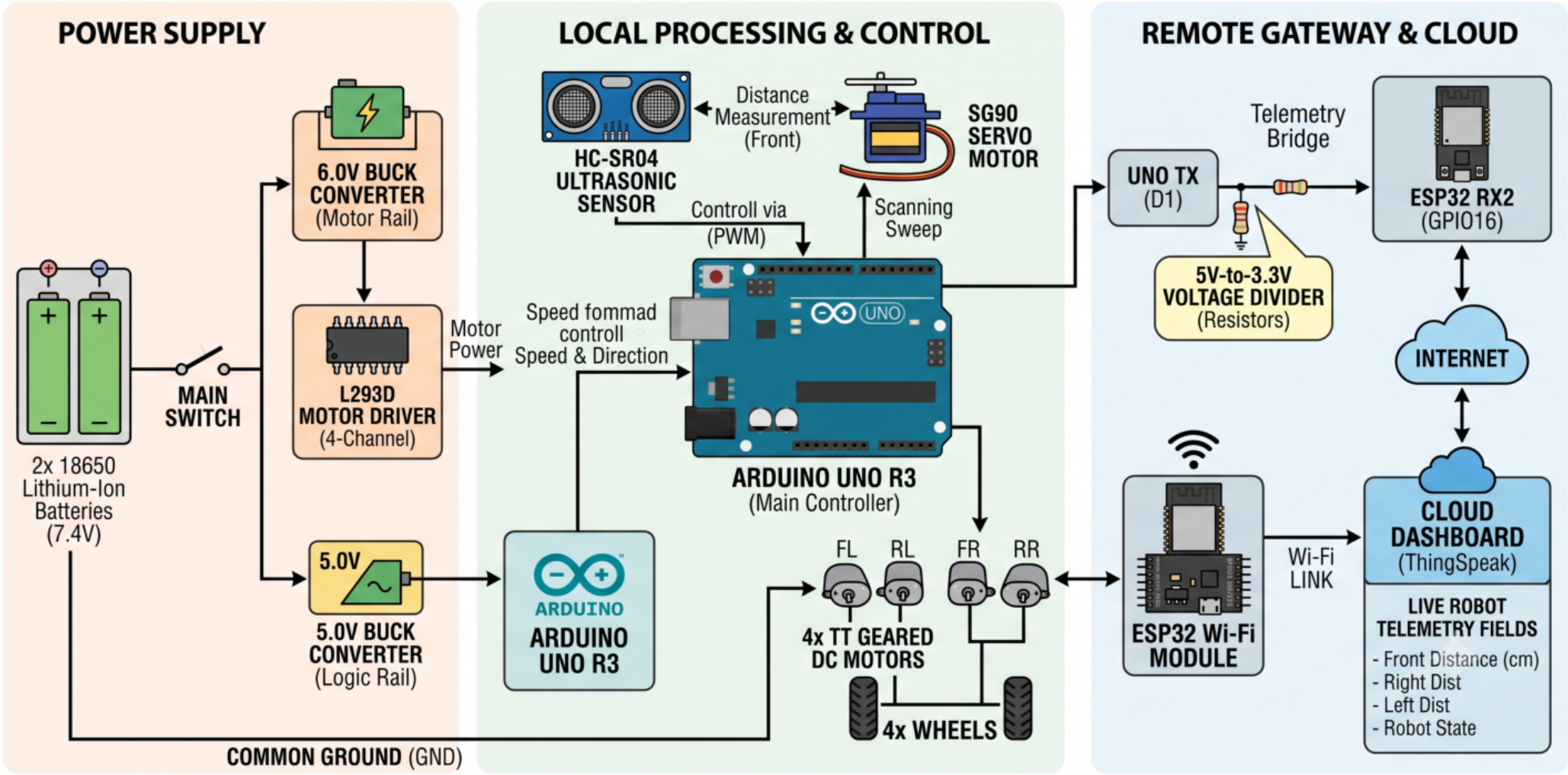
- Offline operation often lacks remote state visibility.
- Fixed-facing sensors cannot compare left/right clearance.
- Complex navigation can exceed the needs of a low-cost 8-bit controller.

Note: full bibliographic details are reproduced from the supplied project report.

HOW THIS PROJECT BUILDS ON THE WORK

- Adds servo-based directional scanning to a reactive loop.
- Separates real-time control (UNO) from Wi-Fi/cloud tasks (ESP32).
- Documents a 1 kΩ / 2 kΩ serial voltage divider and dual power rails.

System Architecture



Major Project Modules

01 Sensing

HC-SR04 measures front clearance.

02 Servo Scan

SG90 sweeps to right and left scan angles.

03 Motor Control

L293D shield drives four TT geared DC motors.

04 Serial Bridge

UNO telemetry passes through a resistor divider to ESP32 RX2.

05 Cloud Telemetry

ESP32 parses fields and posts to ThingSpeak.

06 Dashboard

Shows mode, distances, robot state and controls.

Sensing & Servo Scan

HC-SR04 ULTRASONIC SENSOR

TRIG → UNO A0; ECHO → UNO A1. Measures front distance each control cycle.

SG90 SERVO

Mounted at the front to pan the sensor. Scan positions: 25° right, 155° left, 90° centre.

OBSTACLE DECISION RULE

If front distance is below 30 cm, stop and reverse briefly, then compare left and right clearance.

Distance (cm) = Echo duration (µs) × 0.0343 ÷ 2

A 40 ms gap between pings reduces acoustic echo overlap.

Local Navigation & Motor Control

CONTROL BOARD

Arduino UNO R3 executes the local sense-decide-act loop.

MOTOR DRIVER

Adafruit Motor Shield V1 / L293D drives M1–M4.

DRIVE SYSTEM

Four TT geared DC motors provide forward, reverse and turns.

1 Sense front distance

2 Compare with 30 cm threshold

3 Stop and reverse for 350 ms

4 Scan 25° / 155°

5 Turn toward greater clearance

Configured motor PWM: 170 / 255. Turn delay: approximately 480 ms.

POWER DISTRIBUTION & SERIAL BRIDGE

DUAL POWER RAILS

7.4 V nominal battery pack feeds two buck converters. 6.0 V rail powers motors; 5.0 V rail powers logic and sensors.

UART TELEMETRY BRIDGE

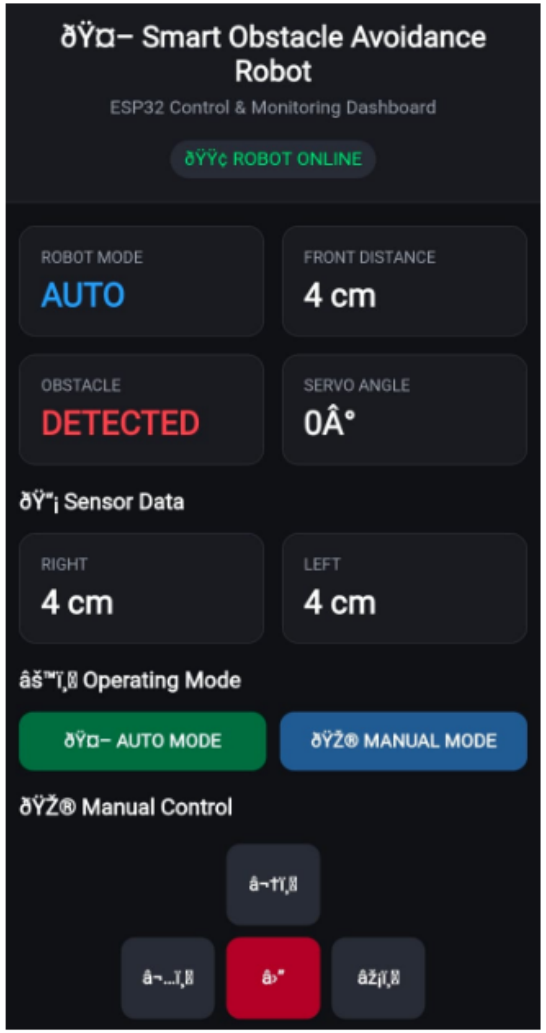
UNO TX (D1) → 1 kΩ / 2 kΩ divider → ESP32 RX2 (GPIO16), at 9600 baud.

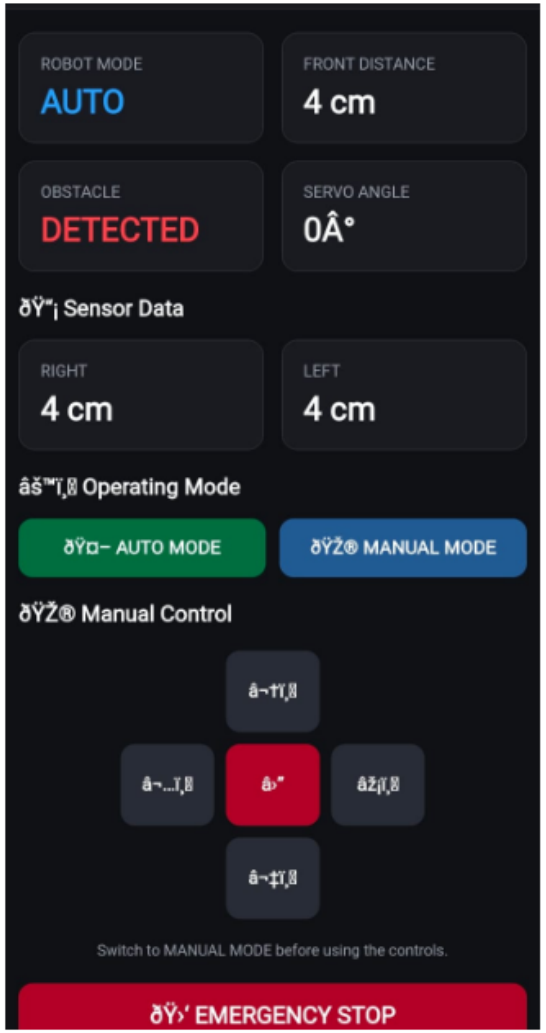
Voltage divider calculation

5.0 V × 2kΩ / (1kΩ + 2kΩ) = 3.33 V

- Common ground across all modules.
- Separate rails reduce motor-noise brownouts.
- Bench check reported 3.31 V peak at ESP32 RX.

Cloud Telemetry & Monitoring Dashboard





- ESP32 receives and parses key-value state lines from the UNO.
- Wi-Fi gateway posts telemetry to ThingSpeak over HTTP.
- Fields include front, left and right distance plus robot state.
- Dashboard supports Auto Mode, Manual Mode and Emergency Stop.
- Cloud updates are configured at a safe interval of 20 seconds.

Example event: 4 cm readings · obstacle detected · AUTO mode

Implementation & Test Verification

- Hardware assembled: 4WD chassis, sensor/servo head and dual power rails.
- UNO and ESP32 firmware uploaded; live serial telemetry verified.
- Dashboard confirmed in Auto and Manual modes, including Emergency Stop.
- Bench tests verified voltage translation and sensor/control integration.
- Floor tests recorded obstacle stop, reverse, scan, turn and cloud status.

TEST	CONDITION	OBSERVED RESULT	STATUS
TC1	Front < 30 cm	Stopped at 28 cm, reversed, scanned and turned left	Verified
TC2	Front > 30 cm	Forward motion; moving_forward telemetry	Verified
TC3	UART level shift	3.31 V peak; no packet loss observed	Verified
TC4	Emergency Stop	Motors halted; E-STOP ACTIVE shown	Verified

Source: Project Report, Table 5.1. Extended floor-trial metrics are shown next.

Verified Parameters & Floor-Test Observations

SAFE DISTANCE

30 cm threshold

TELEMETRY

20-second upload interval

MOTOR SETTING

PWM 170 / 255

Reported indoor floor-test samples

TRIAL	SURFACE	ANGLE	STOP (cm)	TURN ACC.	CLOUD (s)
1	Smooth cardboard	90°	28.4	98.2%	1.12
2	Smooth wood	75°	28.1	96.5%	1.18
3	Acrylic plate	60°	27.6	94.0%	1.05
4	Textured plastic	45°	26.8	91.5%	1.25
5	Soft fabric panel	30°	24.2	85.0%	1.30

Observation: soft fabric and angled surfaces reduced ranging/turn accuracy. These are the report's five recorded samples, not a claim of a universal success rate.

Limitations & Future Enhancements

CURRENT LIMITATIONS

Single ultrasonic sensor can miss narrow or sound-absorbing obstacles; open-loop motors may drift.

NETWORK LIMIT

ThingSpeak updates are limited to approximately one upload per 20 seconds.

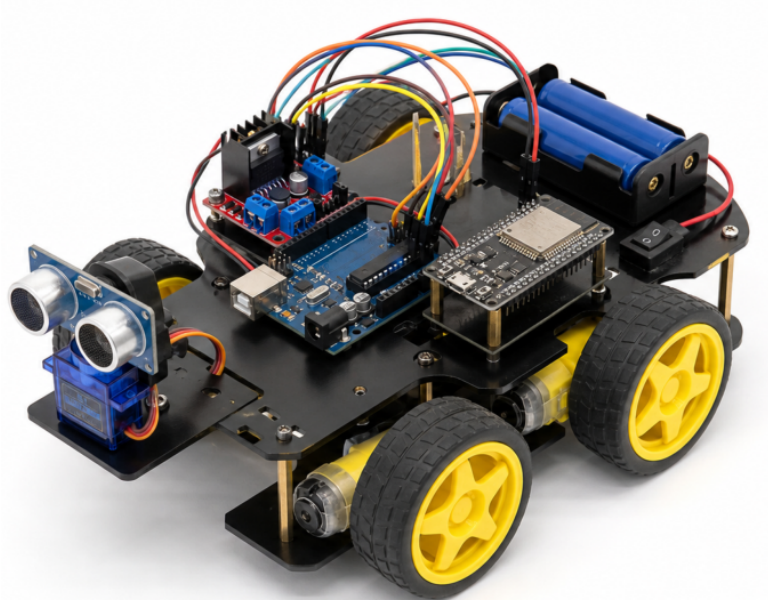
- Add wheel encoders and IMU for closed-loop odometry.
- Consider MQTT for lower-latency scalable telemetry.
- Add ESP32-CAM or LiDAR for richer perception and mapping.
- Add battery-voltage monitoring to the dashboard.
- Extend manual cloud commands with fail-safe handling.
- Test across more obstacle materials, angles and floor conditions.

Current scope remains a low-cost indoor reactive robot, not a SLAM or industrial safety platform.

Conclusion

The project integrates low-cost autonomous obstacle avoidance with remote IoT observability. The Arduino UNO performs deterministic sensing and motor control, while the ESP32 forwards structured telemetry to ThingSpeak over Wi-Fi.

Bench and floor tests documented the key control sequence, safe serial-level translation, dashboard modes and emergency stop. The architecture is modular and suitable for further improvement through odometry, richer sensing and lower-latency messaging.



Key contribution: local safety logic stays on the UNO while the ESP32 provides cloud monitoring.

Thank you · Questions & Feedback

References

[1] R. D. Ganesh, S. S. Kumar, and R. A. Adithya, “IoT-Based Smart Obstacle-Avoiding Robot Using Arduino Uno and ESP32 for Real-Time Remote Monitoring,” Mini Project Report, VISTAS, Chennai, 2026.

[2] K. Katona, H. A. Neamah, and P. Korondi, “Obstacle avoidance and path planning methods for autonomous navigation of mobile robot,” Sensors, vol. 24, no. 11, p. 3573, Jun. 2024.

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[8] MathWorks, “ThingSpeak IoT Analytics Service Documentation,” 2025. https://www.mathworks.com/help/thingspeak/

Reference numbering follows the supplied project report.